

ECE269: Linear Algebra and Applications
Winter 2026
Homework # 4 Solutions

Problem 1

We have the following relations that define $\mathbf{A}^+$ the pseudoinverse of $\mathbf{A}$, which we summarize here.

$$\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}. \quad (1)$$

$$\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+ \quad (2)$$

$$\mathbf{A}^T(\mathbf{A}^+)^T = \mathbf{A}^+\mathbf{A} \quad (3)$$

$$(\mathbf{A}^+)^T\mathbf{A}^T = \mathbf{A}\mathbf{A}^+ \quad (4)$$

a) Let $\mathbf{B}$ and $\mathbf{C}$ be pseudoinverses of $\mathbf{A}$. Then, we have:

$$\mathbf{A}\mathbf{B} \stackrel{(1)}{=} \mathbf{A}\mathbf{C}\mathbf{A}\mathbf{B} \stackrel{(4)}{=} \mathbf{C}^T\mathbf{A}^T\mathbf{B}^T\mathbf{A}^T = \mathbf{C}^T(\mathbf{A}\mathbf{B}\mathbf{A})^T \stackrel{(1)}{=} \mathbf{C}^T\mathbf{A}^T \stackrel{(4)}{=} \mathbf{A}\mathbf{C} \quad (5)$$

Similarly, we can show that $\mathbf{B}\mathbf{A} = \mathbf{C}\mathbf{A}$:

$$\mathbf{B}\mathbf{A} \stackrel{(1)}{=} \mathbf{B}\mathbf{A}\mathbf{C}\mathbf{A} \stackrel{(4)}{=} \mathbf{A}^T\mathbf{B}^T\mathbf{A}^T\mathbf{C}^T = (\mathbf{A}\mathbf{B}\mathbf{A})^T\mathbf{C}^T \stackrel{(1)}{=} \mathbf{A}^T\mathbf{C}^T \stackrel{(4)}{=} \mathbf{C}\mathbf{A} \quad (6)$$

We therefore have

$$\mathbf{B} \stackrel{(2)}{=} \mathbf{B}\mathbf{A}\mathbf{B} \stackrel{(6)}{=} \mathbf{C}\mathbf{A}\mathbf{B} \stackrel{(5)}{=} \mathbf{C}\mathbf{A}\mathbf{C} \stackrel{(2)}{=} \mathbf{C}.$$

b) Since $\mathbf{A}$ is full-rank and tall, $\mathbf{A}^T\mathbf{A}$ is symmetric and non-singular. Defining $\mathbf{B} := (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$, we see that $\mathbf{B} \in \mathbb{R}^{n \times m}$, and

$$\mathbf{A}\mathbf{B}\mathbf{A} = \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A} = \mathbf{A}$$

$$\mathbf{B}\mathbf{A}\mathbf{B} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T = \mathbf{B}$$

$$\mathbf{A}^T\mathbf{B}^T = \mathbf{A}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1} = \mathbf{I}_n = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A} = \mathbf{B}\mathbf{A}$$

$$\mathbf{B}^T\mathbf{A}^T = \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T = \mathbf{A}((\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T) = \mathbf{A}\mathbf{B}.$$

Thus, $(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$ satisfies all 4 properties and is the pseudoinverse of $\mathbf{A}$.

c) Since $\mathbf{A}$ is full-rank and fat, $\mathbf{A}\mathbf{A}^T$ is symmetric and non-singular. Defining $\mathbf{B} := \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}$, we see that $\mathbf{B} \in \mathbb{R}^{n \times m}$, and

$$\mathbf{A}\mathbf{B}\mathbf{A} = \mathbf{A}\mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{A} = \mathbf{A}$$

$$\mathbf{B}\mathbf{A}\mathbf{B} = \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{A}\mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1} = \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1} = \mathbf{B}$$

$$\mathbf{A}^T\mathbf{B}^T = \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{A} = (\mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1})\mathbf{A} = \mathbf{B}\mathbf{A}$$

$$\mathbf{B}^T\mathbf{A}^T = (\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{A}\mathbf{A}^T = \mathbf{I}_m = (\mathbf{A}\mathbf{A}^T)(\mathbf{A}\mathbf{A}^T)^{-1} = \mathbf{A}\mathbf{B}.$$

Thus, $\mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}$ satisfies all 4 properties and is the pseudoinverse of $\mathbf{A}$.

- d) If $\mathbf{A}$ is full-rank and square, then so is $\mathbf{A}^T$, and using part (c) and the uniqueness of the pseudoinverse (proved in part (a)), we can compute the pseudoinverse of $\mathbf{A}$ as

$$\mathbf{A}^+ = \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1} = \mathbf{A}^T(\mathbf{A}^T)^{-1}\mathbf{A}^{-1} = \mathbf{A}^{-1}.$$

- e) Since $\mathbf{A}$ is an (orthogonal) projection matrix, it is symmetric and satisfies $\mathbf{A}^2 = \mathbf{A}$. We then have

$$\mathbf{A} \cdot \mathbf{A} \cdot \mathbf{A} = \mathbf{A}^2 \cdot \mathbf{A} = \mathbf{A} \cdot \mathbf{A} = \mathbf{A}^2 = \mathbf{A}.$$

It also holds that $\mathbf{A}^T\mathbf{A}^T = \mathbf{A}\mathbf{A}$ since $\mathbf{A}$ is symmetric, which show that $\mathbf{A}$ satisfies all 4 properties and is its own pseudoinverse.

- f) Let $\mathbf{B} = (\mathbf{A}^+)^T$. Then $\mathbf{B} \in \mathbb{R}^{m \times n}$, and we have

$$\begin{aligned}\mathbf{A}^T\mathbf{B}\mathbf{A}^T &= \mathbf{A}^T(\mathbf{A}^+)^T\mathbf{A}^T = (\mathbf{A}\mathbf{A}^+\mathbf{A})^T = \mathbf{A}^T \\ \mathbf{B}\mathbf{A}^T\mathbf{B} &= (\mathbf{A}^+)^T\mathbf{A}^T(\mathbf{A}^+)^T = (\mathbf{A}^+\mathbf{A}\mathbf{A}^+)^T = (\mathbf{A}^+)^T = \mathbf{B} \\ (\mathbf{A}^T)^T\mathbf{B}^T &= \mathbf{A}\mathbf{A}^+ = (\mathbf{A}^+)^T\mathbf{A}^T = \mathbf{B}\mathbf{A}^T \\ \mathbf{B}^T(\mathbf{A}^T)^T &= \mathbf{A}^+\mathbf{A} = \mathbf{A}^T(\mathbf{A}^+)^T = \mathbf{A}^T\mathbf{B}.\end{aligned}$$

This shows that $(\mathbf{A}^T)^+ = (\mathbf{A}^+)^T$.

- g) Let $\mathbf{B} := (\mathbf{A}^+)^T\mathbf{A}^+$. Then we have

$$(\mathbf{A}\mathbf{A}^T)\mathbf{B}(\mathbf{A}\mathbf{A}^T) = \mathbf{A}(\mathbf{A}^T(\mathbf{A}^+)^T)\mathbf{A}^+\mathbf{A}\mathbf{A}^T = \mathbf{A}(\mathbf{A}^+\mathbf{A})\mathbf{A}^+\mathbf{A}\mathbf{A}^T = \mathbf{A}\mathbf{A}^+\mathbf{A}\mathbf{A}^T = \mathbf{A}\mathbf{A}^T,$$

and

$$\mathbf{B}(\mathbf{A}\mathbf{A}^T)\mathbf{B} = (\mathbf{A}^+)^T\mathbf{A}^+\mathbf{A}(\mathbf{A}^T(\mathbf{A}^+)^T)\mathbf{A}^+ = (\mathbf{A}^+)^T\mathbf{A}^+\mathbf{A}(\mathbf{A}^+\mathbf{A})\mathbf{A}^+ = (\mathbf{A}^+)^T\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = (\mathbf{A}^+)^T\mathbf{A}^+ = \mathbf{B}.$$

Now we verify the symmetry properties.

$$\mathbf{B}\mathbf{A}\mathbf{A}^T = (\mathbf{A}^+)^T\mathbf{A}^+\mathbf{A}\mathbf{A}^T = (\mathbf{A}^+)^T\mathbf{A}^T(\mathbf{A}^+)^T\mathbf{A}^T = (\mathbf{A}^+\mathbf{A}\mathbf{A}^+)^T\mathbf{A}^T = (\mathbf{A}^+)^T\mathbf{A}^T = \mathbf{A}\mathbf{A}^+,$$

and

$$\mathbf{A}\mathbf{A}^T\mathbf{B} = \mathbf{A}\mathbf{A}^T(\mathbf{A}^+)^T\mathbf{A}^+ = \mathbf{A}\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}\mathbf{A}^+.$$

Thus, $\mathbf{B}$ satisfies all 4 properties and is the pseudoinverse of $\mathbf{A}\mathbf{A}^T$, i.e. $(\mathbf{A}\mathbf{A}^T)^+ = (\mathbf{A}^+)^T\mathbf{A}^+$. Call this relationship (g1). Now, using this relation and part (f), we have

$$(\mathbf{A}^T\mathbf{A})^+ = [\mathbf{A}^T(\mathbf{A}^T)^T]^+ \stackrel{(g1)}{=} ((\mathbf{A}^T)^+)^T(\mathbf{A}^T)^+ \stackrel{(f)}{=} ((\mathbf{A}^+)^T)^T(\mathbf{A}^+)^T = \mathbf{A}^+(\mathbf{A}^+)^T.$$

- h) We will show that $\mathcal{R}(\mathbf{A}^+) \subseteq \mathcal{R}(\mathbf{A}^T)$ and $\mathcal{R}(\mathbf{A}^T) \subseteq \mathcal{R}(\mathbf{A}^+)$. For showing the first part, let $\mathbf{y} \in \mathcal{R}(\mathbf{A}^+)$, then $\mathbf{y} = \mathbf{A}^+\mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^m$. Then, we have

$$\mathbf{y} = \mathbf{A}^+\mathbf{x} = \mathbf{A}^+\mathbf{A}\mathbf{A}^+\mathbf{x} = \mathbf{A}^+\mathbf{A}\mathbf{y} = \mathbf{A}^T(\mathbf{A}^+)^T\mathbf{y}$$

so defining $\tilde{\mathbf{x}} := (\mathbf{A}^+)^T\mathbf{y}$, we see that $\mathbf{y}$ can be written as $\mathbf{A}^T\tilde{\mathbf{x}}$, which shows that $\mathbf{y} \in \mathcal{R}(\mathbf{A}^T)$.

For the opposite direction, if $\mathbf{y} \in \mathcal{R}(\mathbf{A}^T)$ is written as $\mathbf{y} = \mathbf{A}^T \mathbf{x}$, then we can similarly show that

$$\mathbf{y} = \mathbf{A}^T \mathbf{x} = (\mathbf{A}\mathbf{A}^+ \mathbf{A})^T \mathbf{x} = \mathbf{A}^T (\mathbf{A}^+)^T \mathbf{A}^T \mathbf{x} = \mathbf{A}^T (\mathbf{A}^+)^T \mathbf{y} = \mathbf{A}^+ \mathbf{A} \mathbf{y}$$

so $\mathbf{y} = \mathbf{A}^+ \tilde{\mathbf{x}}$, where $\tilde{\mathbf{x}} := \mathbf{A} \mathbf{y}$. Thus, $\mathbf{y} \in \mathcal{R}(\mathbf{A}^+)$.

Therefore, we have shown that $\mathcal{R}(\mathbf{A}^+) = \mathcal{R}(\mathbf{A}^T)$.

Now, let $\mathbf{x} \in \mathcal{N}(\mathbf{A}^+)$. We then have

$$\mathbf{A}^+ \mathbf{x} = 0 \Rightarrow \mathbf{A} \mathbf{A}^+ \mathbf{x} = 0 \Rightarrow (\mathbf{A}^+)^T \mathbf{A}^T \mathbf{x} = 0 \Rightarrow \mathbf{A}^T (\mathbf{A}^+)^T \mathbf{A}^T \mathbf{x} = 0 \Rightarrow (\mathbf{A} \mathbf{A}^+ \mathbf{A})^T \mathbf{x} = 0 \Rightarrow \mathbf{A}^T \mathbf{x} = 0.$$

Thus, $\mathbf{x} \in \mathcal{N}(\mathbf{A}^T)$.

Similarly, if $\mathbf{x} \in \mathcal{N}(\mathbf{A}^T)$, we have

$$\mathbf{A}^T \mathbf{x} = 0 \Rightarrow (\mathbf{A}^+)^T \mathbf{A}^T \mathbf{x} = 0 \Rightarrow \mathbf{A} \mathbf{A}^+ \mathbf{x} = 0 \Rightarrow \mathbf{A}^+ \mathbf{A} \mathbf{A}^+ \mathbf{x} = 0 \Rightarrow \mathbf{A}^+ \mathbf{x} = 0.$$

Thus, $\mathbf{x} \in \mathcal{N}(\mathbf{A}^+)$.

We have therefore shown that $\mathcal{N}(\mathbf{A}^+) = \mathcal{N}(\mathbf{A}^T)$.

i) $\mathbf{P}$ and $\mathbf{Q}$ are symmetric by the properties of $\mathbf{A}^+$, and

$$\mathbf{P}^2 = \mathbf{A} \mathbf{A}^+ \mathbf{A} \mathbf{A}^+ = \mathbf{A} \mathbf{A}^+ = \mathbf{P}.$$

Similarly,

$$\mathbf{Q}^2 = \mathbf{A}^+ \mathbf{A} \mathbf{A}^+ \mathbf{A} = \mathbf{A}^+ \mathbf{A} = \mathbf{Q}.$$

Therefore, $\mathbf{P}$ and $\mathbf{Q}$ are projection matrices.

j) Clearly, for every $\mathbf{x} \in \mathbb{R}^m$, $\mathbf{y} = \mathbf{P} \mathbf{x} = \mathbf{A} \mathbf{A}^+ \mathbf{x} \in \mathcal{R}(\mathbf{A})$. Thus, we are done if we can show that $\mathbf{x} - \mathbf{P} \mathbf{x} \in \mathcal{R}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^T)$ which is $\mathcal{N}(\mathbf{A}^+)$ by (h).

We have

$$\mathbf{A}^+ (\mathbf{x} - \mathbf{P} \mathbf{x}) = \mathbf{A}^+ \mathbf{x} - \mathbf{A}^+ \mathbf{A} \mathbf{A}^+ \mathbf{x} = \mathbf{A}^+ \mathbf{x} - \mathbf{A}^+ \mathbf{x} = 0,$$

therefore $\mathbf{P}$ is indeed the projection onto $\mathcal{R}(\mathbf{A})$.

Similarly, for every $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{z} = \mathbf{Q} \mathbf{x} = \mathbf{A}^+ \mathbf{A} \mathbf{x} \in \mathcal{R}(\mathbf{A}^+) = \mathcal{R}(\mathbf{A}^T)$, where the last equality follows from part (h). Thus, we are done if we can show that $\mathbf{x} - \mathbf{Q} \mathbf{x} \in \mathcal{N}(\mathbf{A})$. We have

$$\mathbf{A} (\mathbf{x} - \mathbf{Q} \mathbf{x}) = \mathbf{A} \mathbf{x} - \mathbf{A} \mathbf{A}^+ \mathbf{A} \mathbf{x} = \mathbf{A} \mathbf{x} - \mathbf{A} \mathbf{x} = 0,$$

therefore $\mathbf{Q}$ is indeed the projection onto $\mathcal{R}(\mathbf{A}^T)$.

k) We have

$$\mathbf{A}^+ (\mathbf{A} \mathbf{x}^* - \mathbf{b}) = \mathbf{A}^+ \mathbf{A} \mathbf{A}^+ \mathbf{b} - \mathbf{A}^+ \mathbf{b} = \mathbf{A}^+ \mathbf{b} - \mathbf{A}^+ \mathbf{b} = 0.$$

Hence $\mathbf{A} \mathbf{x}^* - \mathbf{b}$ is orthogonal to $\mathcal{R}((\mathbf{A}^+)^T) = \mathcal{R}((\mathbf{A}^T)^+)$ which is $\mathcal{R}(\mathbf{A})$ by (h) and by orthogonality principle, $\mathbf{x}^*$ is indeed a least-squares solution to $\mathbf{A} \mathbf{x} = \mathbf{b}$.

- l) Suppose that the linear equation $\mathbf{b} = \mathbf{A}\mathbf{x}$ has a solution $\tilde{\mathbf{x}}$. Then, $\mathbf{A}\mathbf{x}^* = \mathbf{A}\mathbf{A}^+\mathbf{b} = \mathbf{A}\mathbf{A}^+\mathbf{A}\tilde{\mathbf{x}} = \mathbf{A}\tilde{\mathbf{x}} = \mathbf{b}$. Now, let $\mathbf{z}$ be any other solution to $\mathbf{b} = \mathbf{A}\mathbf{x}$, i.e., we have $\mathbf{A}\mathbf{z} = \mathbf{b}$.

Then $(\mathbf{z} - \mathbf{x}^*) \in \mathcal{N}(\mathbf{A})$. Also $\mathbf{x}^* \in \mathcal{R}(\mathbf{A}^+) = \mathcal{R}(\mathbf{A}^T)$.

Since $\mathcal{N}(\mathbf{A}) \perp \mathcal{R}(\mathbf{A}^T)$,

$$(\mathbf{x}^*)^T(\mathbf{z} - \mathbf{x}^*) = 0$$

We then have

$$\|\mathbf{z}\|^2 = \|\mathbf{x}^* + (\mathbf{z} - \mathbf{x}^*)\|^2 = \|\mathbf{x}^*\|^2 + \|\mathbf{z} - \mathbf{x}^*\|^2 \geq \|\mathbf{x}^*\|^2.$$

Problem 2

- a) Consider the characteristic polynomial of $\mathbf{A}$, namely, $X_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I})$ ¹. Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of $\mathbf{A}$ (note that they can be repeated with respect to their algebraic multiplicity).

Since these eigenvalues (repeated according to their algebraic multiplicities) will be the roots of the characteristic polynomial $X_{\mathbf{A}}(\lambda)$, therefore, we can also write that,

$$X_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = \prod_{i=1}^n (\lambda_i - \lambda)$$

Now, if we evaluate the above expression at $\lambda = 0$, we find that,

$$X_{\mathbf{A}}(0) = \det(\mathbf{A}) = \lambda_1 \lambda_2 \dots \lambda_n$$

Thus, we finally prove that, $\det(\mathbf{A}) = \lambda_1 \lambda_2 \dots \lambda_n$

- b) We have

$$X_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = \prod_{i=1}^n (\lambda_i - \lambda)$$

It is straightforward to show that coefficient of λ^{n-1} in $X_{\mathbf{A}}(\lambda)$ is $(-1)^{n-1}(\lambda_1 + \lambda_2 + \dots + \lambda_n)$.

Now we show that this coefficient is also equal to $(-1)^{n-1} \text{tr}(\mathbf{A})$.

Consider the definition of $\det(\mathbf{A} - \lambda\mathbf{I})$ (expansion along first row),

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (A_{11} - \lambda) \det(\mathbf{M}_{11}) - A_{12} \det(\mathbf{M}_{12}) + A_{13} \det(\mathbf{M}_{13}) + \dots + (-1)^{n+1} A_{1n} \det(\mathbf{M}_{1n})$$

Here, $\mathbf{M}_{1i}$ is a co-factor corresponding to A_{1i} which is a matrix obtained from $\mathbf{A}$ by deleting row 1 and column i . The only term in this expansion that can generate λ^{n-1} is the first term (i.e., $(A_{11} - \lambda) \det(\mathbf{M}_{11})$). Since in the other terms $((A_{1i} - \lambda) \det(\mathbf{M}_{1i}))$, for $i > 1$, $\mathbf{M}_{1i}$ is given by removing first row and i^{th} column ($i > 1$) of $\mathbf{A}$, this will remove two factors of

¹Note that $\det(\lambda\mathbf{I} - \mathbf{A}) = (-1)^n \det(\mathbf{A} - \lambda\mathbf{I})$, which implies that $\det(\lambda\mathbf{I} - \mathbf{A})$ and $\det(\mathbf{A} - \lambda\mathbf{I})$ have same roots. Since we are only interested in the roots of characteristic polynomial we consider $\det(\mathbf{A} - \lambda\mathbf{I})$ as characteristic polynomial of $\mathbf{A}$

λ (one in $[\mathbf{A} - \lambda \mathbf{I}]_{11}$, and one in $[\mathbf{A} - \lambda \mathbf{I}]_{ii}$), and the maximum power of λ in these terms is $n - 2$. Let call sum of these terms as $f_0(\lambda)$:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = (A_{11} - \lambda) \det(\mathbf{M}_{11}) + f_0(\lambda) \quad (7)$$

Now let focus on $\det(\mathbf{M}_{11})$, which is given by

$$\det(\mathbf{M}_{11}) = (A_{22} - \lambda) \det(\mathbf{M}_{22}^{(1)}) + g(\lambda) \quad (8)$$

where $\mathbf{M}_{22}^{(1)}$ is a matrix obtained from $\mathbf{A}$ by deleting rows 1, 2 and columns 1, 2. With the similar argument, we can show that the maximum power of λ in $g(\lambda)$ is $n - 3$. By replacing 8 in 7, we have

$$\begin{aligned} \det(\mathbf{A} - \lambda \mathbf{I}) &= (A_{11} - \lambda)((A_{22} - \lambda) \det(\mathbf{M}_{22}^{(1)}) + g(\lambda)) + f_0(\lambda) \\ &= (A_{11} - \lambda)(A_{22} - \lambda) \det(\mathbf{M}_{22}^{(1)}) + \underbrace{(A_{11} - \lambda)g(\lambda) + f_0(\lambda)}_{f_1(\lambda)} \end{aligned}$$

The maximum power of λ in $f_1(\lambda)$ is $n - 2$ and only first term can generate λ^{n-1} . By continuing this argument, we get

$$\det(\mathbf{A} - \lambda \mathbf{I}) = (A_{11} - \lambda)(A_{22} - \lambda)(A_{33} - \lambda) \cdots (A_{nn} - \lambda) + f_n(\lambda)$$

where the maximum power of λ in $f_n(\lambda)$ is $n - 2$. Therefore, if we expand the first product term in the above expression, you will find that the coefficient of λ^{n-1} in $\det(\mathbf{A} - \lambda \mathbf{I})$ is given by $(-1)^{n-1}(A_{11} + A_{22} + A_{33} + \cdots + A_{nn}) = (-1)^{n-1} \text{tr}(\mathbf{A})$. Hence,

$$\text{tr}(\mathbf{A}) = \lambda_1 + \lambda_2 + \dots + \lambda_n$$

c) For any $\lambda \in \mathbb{C}$, let $\mu_1, \mu_2, \dots, \mu_k$ be roots of $p(x) = x^k - \lambda = 0$ ², then we have

$$x^k - \lambda = \prod_{i=1}^k (x - \mu_i) = \sum_{l=0}^k c_l x^{k-l} \quad (9)$$

where $c_0 = 1$, and for $1 \leq l \leq k$, $|c_l|$ is the sum of all possible products of the μ_j 's taken l at a time. For example, $c_k = (-1)^k \prod_{i=1}^k \mu_i$. By equating the coefficients of like powers of x in both sides of 9, we conclude that

$$\begin{cases} c_0 = 1 \\ c_l = 0 & l = 1, 2, \dots, k-1 \\ c_k = -\lambda \end{cases} \quad (10)$$

Now, we form the following polynomial of matrix $\mathbf{A}$, with same μ_i 's as defined above,

$$p(\mathbf{A}) = \prod_{i=1}^k (\mathbf{A} - \mu_i \mathbf{I}) = c_k \mathbf{I} + \sum_{l=0}^{k-1} c_l \mathbf{A}^{k-l} \quad (11)$$

²note that in this case μ_i 's are k^{th} roots of λ , i.e., $\mu_i = |\lambda|^{\frac{1}{k}} e^{j \frac{2\pi i}{k}}$

By replacing 10 in 11 we get

$$p(\mathbf{A}) = \mathbf{A}^k - \lambda \mathbf{I} \quad (12)$$

The characteristic polynomial of $\mathbf{A}$ is given by,

$$\mathbf{X}_A(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I}) = \prod_{j=1}^n (\lambda_j - \lambda) \quad (13)$$

Now consider the characteristic polynomial of $\mathbf{A}^k$, we have

$$\begin{aligned} \mathbf{X}_{A^k}(\lambda) &= \det(\mathbf{A}^k - \lambda \mathbf{I}) \stackrel{(1)}{=} \det \left(\prod_{i=1}^k (\mathbf{A} - \mu_i \mathbf{I}) \right) \stackrel{(2)}{=} \prod_{i=1}^k \det(\mathbf{A} - \mu_i \mathbf{I}) \\ &\stackrel{(3)}{=} \prod_{i=1}^k \prod_{j=1}^n (\lambda_j - \mu_i) = \prod_{j=1}^n \prod_{i=1}^k (\lambda_j - \mu_i) \stackrel{(4)}{=} \prod_{j=1}^n (\lambda_j^k - \lambda) \end{aligned} \quad (14)$$

which shows that the eigenvalues of $\mathbf{A}^k$ are exactly $\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k$.

In above derivation, (1) is correct since we replaced 12 with 11, (2) is correct since determinant of product of matrices is equal to product of determinant of matrices. In (3) we used equation 13 where λ is replaced by μ_i , in (4) we used equation 9 where x is replaced by λ_j . (Proof is complete at this point, the rest is an alternative argument which **cannot** be considered as a complete solution to this question)

Let assume $\mathbf{v}_i$ is corresponding eigenvector of λ_i . We can use the following argument to show that $\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k$ are eigenvalues of $\mathbf{A}^k$:

$$\begin{aligned} \mathbf{A} \mathbf{v}_i &= \lambda_i \mathbf{v}_i \\ \mathbf{A}^2 \mathbf{v}_i &= \mathbf{A} \mathbf{A} \mathbf{v}_i = \lambda_i \mathbf{A} \mathbf{v}_i = \lambda_i^2 \mathbf{v}_i \\ \mathbf{A}^3 \mathbf{v}_i &= \mathbf{A} \mathbf{A}^2 \mathbf{v}_i = \lambda_i^2 \mathbf{A} \mathbf{v}_i = \lambda_i^3 \mathbf{v}_i \\ &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ &\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ \mathbf{A}^k \mathbf{v}_i &= \mathbf{A} \mathbf{A}^{k-1} \mathbf{v}_i = \lambda_i^{k-1} \mathbf{A} \mathbf{v}_i = \lambda_i^k \mathbf{v}_i \end{aligned}$$

However, this argument cannot guarantee that $\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k$ are the only eigenvalues and $\mathbf{A}^k$ does not have any other eigenvalue out of this set, whereas the previous proof finds the roots of characteristic polynomial of $\mathbf{A}^k$ and this ensures $\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k$ are the only eigenvalues of $\mathbf{A}^k$.

d) Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of $\mathbf{A}$. We have:

$$\mathbf{A} \text{ is invertible} \iff \det(\mathbf{A}) \neq 0 \iff \prod_{i=1}^n \lambda_i \neq 0 \iff \lambda_i \neq 0 \forall i$$

e) If $\mathbf{A}$ is invertible, we know that $\lambda_i \neq 0$ for all i . We have

$$\begin{aligned}
X_{\mathbf{A}^{-1}}(\lambda) &= \det(\mathbf{A}^{-1} - \lambda \mathbf{I}) \\
&= \det(\mathbf{A}^{-1}(\mathbf{I} - \lambda \mathbf{A})) \\
&= \det(\mathbf{I} - \lambda \mathbf{A}) \det(\mathbf{A}^{-1}) \\
&= \lambda^n \det(\lambda^{-1} \mathbf{I} - \mathbf{A}) \det(\mathbf{A})^{-1} \\
&= (-\lambda)^n \det(\mathbf{A} - \lambda^{-1} \mathbf{I}) \det(\mathbf{A})^{-1} \\
&= (-\lambda)^n X_{\mathbf{A}}(\lambda^{-1}) \det(\mathbf{A})^{-1} \\
&= (-1)^n X_{\mathbf{A}}(\lambda^{-1}) \prod_{i=1}^n \frac{\lambda}{\lambda_i} \\
&= (-1)^n \prod_{i=1}^n \frac{\lambda}{\lambda_i} (\lambda_i - \lambda^{-1}) \\
&= \prod_{i=1}^n (\lambda_i^{-1} - \lambda)
\end{aligned}$$

which shows that $\lambda_1^{-1}, \lambda_2^{-1}, \dots, \lambda_n^{-1}$ are the eigenvalues of $\mathbf{A}^{-1}$

Problem 3

- (a) Suppose $\mathbf{A}$ has a non-zero eigenvalue λ . Then by Problem 2, part (c), λ^k is an eigenvalue of $\mathbf{A}^k$. However, $\mathbf{A}^k = \mathbf{0}$ and hence all its eigenvalues are zero [Can also verify this quickly by observing the characteristic polynomial $\det(\mathbf{A}^k - \lambda \mathbf{I}) = \det(-\lambda \mathbf{I})$.] Therefore, all eigenvalues of $\mathbf{A}$ must be zero.
- (b) Firstly, convince yourself that for any $i \geq 1$, $\mathcal{N}(\mathbf{A}^i) \subseteq \mathcal{N}(\mathbf{A}^{i+1})$. Now suppose that for some integer p , $\mathcal{N}(\mathbf{A}^p) = \mathcal{N}(\mathbf{A}^{p+1})$. Consider a vector $\mathbf{x} \in \mathcal{N}(\mathbf{A}^{p+2})$. Then $\mathbf{A}^{p+2}\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{A}\mathbf{x} \in \mathcal{N}(\mathbf{A}^{p+1}) \Rightarrow \mathbf{A}\mathbf{x} \in \mathcal{N}(\mathbf{A}^p) \Rightarrow \mathbf{A}^p\mathbf{A}\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} \in \mathcal{N}(\mathbf{A}^{p+1})$. Hence, we have $\mathcal{N}(\mathbf{A}^{p+2}) = \mathcal{N}(\mathbf{A}^{p+1}) = \mathcal{N}(\mathbf{A}^p)$. It is easy to see that this argument can be extended to show that $\mathcal{N}(\mathbf{A}^{p+l}) = \mathcal{N}(\mathbf{A}^p)$ for all $l \geq 1$.

Returning to our problem, k is the smallest integer such that $\mathbf{A}^k = \mathbf{0}$. Hence $\mathbf{A}^i \neq \mathbf{0}$ for $i < k$. We claim that we must have the following chain of *strict inclusions*

$$\mathcal{N}(\mathbf{A}^i) \subset \mathcal{N}(\mathbf{A}^{i+1}), \quad 1 \leq i \leq k-1.$$

Otherwise, if $\mathcal{N}(\mathbf{A}^{i_0}) = \mathcal{N}(\mathbf{A}^{i_0+1})$ for some $i_0 \leq k-1$, then by the previous argument, $\mathcal{N}(\mathbf{A}^{i_0}) = \mathcal{N}(\mathbf{A}^k)$, which is not possible since $\mathbf{A}^k = \mathbf{0}$ and $\mathbf{A}^{i_0} \neq \mathbf{0}$. Hence, the dimensions of the null-spaces strictly increase in each step of the above inclusion. Moreover $\dim(\mathcal{N}(\mathbf{A})) \geq 1$, since $\mathbf{A}$ has zero eigenvalues. Hence $\dim(\mathcal{N}(\mathbf{A}^k)) \geq k$. But the null space of $\mathbf{A}^k$ is $\mathbb{C}^n$. Hence $k \leq n$.

- (c) First convince yourself that $\mathbf{A}^i \mathbf{x} \neq \mathbf{0}$ for all $i \leq n-1$. Let

$$\sum_{i=0}^{n-1} \alpha_i \mathbf{A}^i \mathbf{x} = \mathbf{0},$$

where $\mathbf{A}^0 = \mathbf{I}$. Multiply by $\mathbf{A}^{n-1}$ on both sides to obtain $\alpha_0 \mathbf{A}^{n-1} \mathbf{x} = \mathbf{0}$. Since $\mathbf{A}^{n-1} \mathbf{x} \neq \mathbf{0}$, this implies that $\alpha_0 = 0$. Similarly, by multiplying with $\mathbf{A}^{n-2}$, argue that $\alpha_1 = 0$. Continue this procedure to complete the proof (DIY).

Problem 4

- (a) $\mathbf{A}^H \mathbf{A}$ is a Hermitian ($(\mathbf{A}^H \mathbf{A})^H = \mathbf{A}^H \mathbf{A}$) and PSD matrix ($(\mathbf{x}^H \mathbf{A}^H \mathbf{A} \mathbf{x}) = \|\mathbf{A} \mathbf{x}\|^2 \geq 0$ for all $\mathbf{x}$), thus its singular values are the same as its eigenvalues.

Proof: Let define $\mathbf{B} = \mathbf{A}^H \mathbf{A}$, we have

$$\sigma_i(\mathbf{B}) = \sqrt{\lambda_i(\mathbf{B}^H \mathbf{B})} = \sqrt{\lambda_i(\mathbf{B}^2)} \stackrel{(a)}{=} \sqrt{\lambda_i^2(\mathbf{B})} = \lambda_i(\mathbf{B}) \quad (15)$$

where (a) is from Problem 2 part c. Also we used the fact that all eigenvalues of a symmetric PSD matrix are real and non-negative (if it was not PSD then its singular values were the same as absolute value of its eigenvalues).

End

Using this we can write that

$$\|\mathbf{A}^H \mathbf{A}\| = \sigma_1 = \lambda_1$$

where σ_1 is largest largest singular value and λ_1 is largest eigenvalue of $\mathbf{A}^H \mathbf{A}$. By definition of spectral norm we have,

$$\|\mathbf{A}\|^2 = \left(\sqrt{\lambda_{\max}(\mathbf{A}^H \mathbf{A})} \right)^2 = \lambda_{\max}(\mathbf{A}^H \mathbf{A}) = \lambda_1$$

- (b) Since $\|\mathbf{U} \mathbf{x}\| = \|\mathbf{x}\|$ for every unitary matrix $\mathbf{U}$, $\|\mathbf{U} \mathbf{A} \mathbf{V}\| = \max_{\mathbf{x} \neq 0} \frac{\|\mathbf{U} \mathbf{A} \mathbf{V} \mathbf{x}\|}{\|\mathbf{x}\|} = \max_{\mathbf{x} \neq 0} \frac{\|\mathbf{A} \mathbf{V} \mathbf{x}\|}{\|\mathbf{x}\|} = \max_{\mathbf{x} \neq 0} \frac{\|\mathbf{A} \mathbf{V} \mathbf{x}\|}{\|\mathbf{V} \mathbf{x}\|}$. Since $\mathbf{V}$ is a unitary transformation, $\{\mathbf{x} | \mathbf{x} \neq 0\} = \{\mathbf{x} | \mathbf{V} \mathbf{x} \neq 0\}$. Substituting $\mathbf{V} \mathbf{x} = \mathbf{y}$ yields $\max_{\mathbf{x} \neq 0} \frac{\|\mathbf{A} \mathbf{V} \mathbf{x}\|}{\|\mathbf{V} \mathbf{x}\|} = \max_{\mathbf{y} \neq 0} \frac{\|\mathbf{A} \mathbf{y}\|}{\|\mathbf{y}\|} = \|\mathbf{A}\|$.

- (c) If $\mathbf{U}_A \Sigma_A \mathbf{V}_A^H$ is an SVD of $\mathbf{A}$, and $\mathbf{U}_B \Sigma_B \mathbf{V}_B^H$ is an SVD of $\mathbf{B}$, $\begin{bmatrix} \mathbf{U}_A & \mathbf{0} \\ \mathbf{0} & \mathbf{U}_B \end{bmatrix} \begin{bmatrix} \Sigma_A & \mathbf{0} \\ \mathbf{0} & \Sigma_B \end{bmatrix} \begin{bmatrix} \mathbf{V}_A^H & \mathbf{0} \\ \mathbf{0} & \mathbf{V}_B^H \end{bmatrix}$ is an SVD of $\begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{bmatrix}$. This shows that the singular values of $\begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{bmatrix}$ are the union of singular values of $\mathbf{A}$ and $\mathbf{B}$ (including multiplicity), which in turn implies that $\left\| \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{bmatrix} \right\| = \max(\|\mathbf{A}\|, \|\mathbf{B}\|)$.

Problem 5

- a) Since $\mathbf{A}$ is symmetric and PSD matrix we can write it as $\mathbf{A} = \sum_{i=1}^n \lambda_i \mathbf{u}_i \mathbf{u}_i^T$ with $\lambda_i \geq 0$.

$$\text{tr}(\mathbf{A} \mathbf{B}) = \text{tr} \left(\sum_{i=1}^n \lambda_i \mathbf{u}_i \mathbf{u}_i^T \mathbf{B} \right) = \sum_{i=1}^n \lambda_i \text{tr}(\mathbf{u}_i \mathbf{u}_i^T \mathbf{B}) = \sum_{i=1}^n \lambda_i \text{tr}(\mathbf{u}_i^T \mathbf{B} \mathbf{u}_i) \geq 0 \quad (16)$$

Note that we used properties of trace given in discussion-7, also since $\mathbf{B}$ is symmetric and PSD matrix, we have $\mathbf{u}_i^T \mathbf{B} \mathbf{u}_i \geq 0$.

Another Solution

Since $\mathbf{A}$ is symmetric and PSD, in class we learned that, we can say $\mathbf{A} = \mathbf{U} \Lambda_{\mathbf{A}} \mathbf{U}^T = \mathbf{U} \Lambda_{\mathbf{A}}^{1/2} \Lambda_{\mathbf{A}}^{1/2} \mathbf{U}^T = \mathbf{U} \Lambda_{\mathbf{A}}^{1/2} (\mathbf{U} \Lambda_{\mathbf{A}}^{1/2})^T$, let define $\tilde{\mathbf{U}} := \mathbf{U} \Lambda_{\mathbf{A}}^{1/2}$. We have $\mathbf{A} = \tilde{\mathbf{U}} \tilde{\mathbf{U}}^T$. Similarly, we can write $\mathbf{B} = \tilde{\mathbf{V}} \tilde{\mathbf{V}}^T$. We then have $\text{trace}(\mathbf{A} \mathbf{B}) = \text{trace}((\tilde{\mathbf{U}} \tilde{\mathbf{U}}^T \tilde{\mathbf{V}}) \tilde{\mathbf{V}}^T) = \text{trace}(\tilde{\mathbf{V}}^T (\tilde{\mathbf{U}} \tilde{\mathbf{U}}^T \tilde{\mathbf{V}}))$. Since $\tilde{\mathbf{V}}^T \tilde{\mathbf{U}} \tilde{\mathbf{U}}^T \tilde{\mathbf{V}} = (\tilde{\mathbf{U}}^T \tilde{\mathbf{V}})^T (\tilde{\mathbf{U}}^T \tilde{\mathbf{V}})$ is symmetric, positive semi-definite, its trace (which from problem 2 part b we know it is the sum of the eigenvalues) is non-negative.

- b) Since $\mathbf{A} = (\mathbf{A} + \mathbf{B}) - \mathbf{B} \succeq 0$, we have $\mathbf{A} + \mathbf{B} \succeq \mathbf{B}$.
- c) If $\mathbf{A} \succeq \mathbf{B}$, we have $\mathbf{A} - \mathbf{B} \succeq 0$, which implies that $-\mathbf{B} - (-\mathbf{A}) \succeq 0$, and thus that $-\mathbf{B} \succeq -\mathbf{A}$.
- d) Since $\mathbf{A}$ is symmetric, we have $\mathbf{A} = \mathbf{Q} \Lambda \mathbf{Q}^T$, where $\mathbf{Q} \mathbf{Q}^T = \mathbf{I}$. Thus, $\mathbf{A} - \mathbf{I} = \mathbf{Q} (\Lambda - \mathbf{I}) \mathbf{Q}^T$, and the eigenvalues of $\mathbf{A} - \mathbf{I}$ are the eigenvalues of $\mathbf{A}$ minus 1. Thus, if $\mathbf{A} - \mathbf{I}$ is positive semi-definite, every eigenvalue λ of $\mathbf{A} \geq 1$. Now, we have $\mathbf{A}^{-1} = \mathbf{Q} \Lambda^{-1} \mathbf{Q}^T$ with eigenvalues ≤ 1 . Hence, all eigenvalues of $\mathbf{I} - \mathbf{A}^{-1} = \mathbf{Q} (\mathbf{I} - \Lambda^{-1}) \mathbf{Q}^T$ are non-negative, thus $\mathbf{I} - \mathbf{A}^{-1}$ is positive semi-definite, which means $\mathbf{I} \succeq \mathbf{A}^{-1}$.
- e) Since $\mathbf{B} = \mathbf{Q} \Lambda \mathbf{Q}^T \succ 0$, $\mathbf{B}^{1/2} = \mathbf{Q} \Lambda^{1/2} \mathbf{Q}^T \succ 0$. Then by part (a), $\mathbf{A} - \mathbf{B} \succeq 0$ implies $\mathbf{B}^{-1/2} (\mathbf{A} - \mathbf{B}) \mathbf{B}^{-1/2} = \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} - \mathbf{I} \succeq 0$. Hence by part (d), $\mathbf{I} - \mathbf{B}^{1/2} \mathbf{A}^{-1} \mathbf{B}^{1/2} \succeq 0$. Finally, by part (a) once again, $\mathbf{B}^{-1/2} (\mathbf{I} - \mathbf{B}^{1/2} \mathbf{A}^{-1} \mathbf{B}^{1/2}) \mathbf{B}^{-1/2} = \mathbf{B}^{-1} - \mathbf{A}^{-1} \succeq 0$.